



NAME: _____

TEACHER: _____

GOSFORD HIGH SCHOOL

2012

EXTENSION 2 MATHEMATICS HSC ASSESSMENT TASK 3.

Time Allowed: 60 minutes (plus 5 min. reading time)

- Write using black or blue pen.
- Board-approved calculators may be used.
- Each Section should be started on a new page.
- All necessary working should be shown in Section II and Section III.

SPECIAL INSTRUCTIONS: Tear off back page (multiple choice/standard integrals)

SECTION	QUESTION TYPE	MARKS	RESULT
I	MULTIPLE CHOICE	4	
II	EXTENDED RESPONSE	18	
III	EXTENDED RESPONSE	18	
	TOTAL	40	

SECTION I: MULTIPLE CHOICE. (4 MARKS)

1. The equation $x^2 + (2 - r^2)y^2 = 1$ represents a conic section. For what values of r is the curve a hyperbola

A. $r = \pm 1$. B. $r = \pm\sqrt{2}$. C. $-\sqrt{2} < r < \sqrt{2}$. D. $r < -\sqrt{2}$ or $r > \sqrt{2}$.

2. The eccentricity of $\frac{x^2}{4} - \frac{y^2}{3} = 1$ is

A. $\frac{1}{2}$. B. $\frac{1}{4}$ C. $\frac{\sqrt{7}}{2}$ D. $\frac{7}{4}$

3. If $\frac{1}{1-x^2} = \frac{A}{1-x} + \frac{B}{1+x}$ then

A. $A = 1, B = 1$ B. $A = \frac{1}{2}, B = \frac{1}{2}$ C. $A = 2, B = 2$ D. $A = -1, B = 2$

4. $\int \sin^{-1}x \, dx =$

A. $x \sin^{-1}x - \int \frac{x}{\sqrt{1-x^2}} \, dx$ C. $\frac{1}{\sqrt{1-x^2}} - \int x \sin^{-1}x \, dx$
B. $\frac{\sin^{-1}x}{\sqrt{1-x^2}} - \int x \, dx$ D. $\frac{x}{\sqrt{1-x^2}} - \int \sin^{-1}x \, dx$

SECTION II: (18 MARKS) Show all necessary working.

1. Consider the ellipse $\mathcal{E}$ with equation $3x^2 + 4y^2 = 12$.

(i) Calculate the eccentricity of $\mathcal{E}$. (1)

(ii) Find the coordinates of the foci S and S' of $\mathcal{E}$. (1)

(iii) Find the equations of the directrices of $\mathcal{E}$. (1)

(iv) Show that the equation of the tangent at the point $P_0(x_0, y_0)$ on $\mathcal{E}$ is
$$\frac{xx_0}{4} + \frac{yy_0}{3} = 1$$
 (3)

(v) If P is an arbitrary point on $\mathcal{E}$ prove that the sum of the distances SP and $S'P$ is a constant. (2)

2. The point $P(a \sec \theta, b \tan \theta)$ lies on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

(i) Show that the equation of the normal at P is given by
 $a \tan \theta x + b \sec \theta y = (a^2 + b^2) \sec \theta \tan \theta$. (3)

(ii) If the normal at P meets the x-axis at G and N is the foot of the perpendicular from P to the x-axis show that $NG : ON = b^2 : a^2$ where O is the origin. (3)

3. Let $P(ct, \frac{c}{t})$, where $t > 0$, be any point on the rectangular hyperbola $xy = c^2$. The tangent at P meets the x-axis at Q . Show that the locus of the midpoint M of PQ is another rectangular hyperbola. (4)

SECTION III: (18 MARKS) **START A NEW PAGE.** Show all necessary working.

1. (a) Evaluate $\int_0^2 \frac{2}{\sqrt{x^2+16}} dx$. (1)

(b) Find $\int \frac{4}{x^2-4x+3} dx$. (4)

(c) Show that $\int_4^{12} \frac{dx}{\sqrt{x}(4+x)} = \frac{\pi}{12}$. (4)

(d) Use the substitution $t = \tan \frac{x}{2}$ to show that $\int \frac{dx}{1+\sin x} = 2 \int \frac{dt}{(1+t)^2}$. (3)

(e) Use integration by parts to find $\int x e^{3x} dx$. (3)

2. The equation of a curve is $x^2 + xy + 4y^2 = 4$. Use implicit differentiation to find the equation of the tangent to the curve at the point $(1, -1)$ on it. (3)

Name: _____

Teacher: _____

Multiple-choice answer sheet

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely, using a black pen.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
 correct
 ↓

Start
here →

1. A ☐ B ☐ C ☐ D ☐

2. A ☐ B ☐ C ☐ D ☐

3. A ☐ B ☐ C ☐ D ☐

4. A ☐ B ☐ C ☐ D ☐

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x$, $x > 0$

SOLUTIONS

SECTION I

1. D
2. C
3. B
4. A

SECTION II

1. (i) If $3x^2 + 4y^2 = 12$

$$\frac{3x^2}{12} + \frac{4y^2}{12} = \frac{12}{12}$$

$$\text{i.e. } \frac{x^2}{4} + \frac{y^2}{3} = 1$$

Now $b^2 = a^2(1 - e^2)$

$$3 = 4(1 - e^2)$$

$$3 = 4 - 4e^2$$

$$4e^2 = 1$$

$$e^2 = \frac{1}{4}$$

$$e = \frac{1}{2}$$

(1)

(ii) Foci are $(\pm ae, 0)$

$$ae = 2 \times \frac{1}{2}$$

$$= 1$$

$$\therefore S \text{ is } (1, 0), S' \text{ is } (-1, 0)$$

(1)

(iii) Directrices are $x = \pm \frac{a}{e}$

$$\frac{a}{e} = \frac{2}{\frac{1}{2}}$$

$$= 4$$

$$\therefore \text{Directrices are } x = \pm 4$$

(1)

(iv) Given $\frac{x^2}{4} + \frac{y^2}{3} = 1$

Diff. w.r.t x

$$\frac{2x}{4} + \frac{2y}{3} \cdot y' = 0$$

$$\frac{2y}{3} \cdot y' = -\frac{x}{2}$$

$$\frac{y'}{y} = \frac{-x/2}{2y/3}$$

$$= \frac{-3x}{4y}$$

At $P(x_0, y_0)$ $y' = \frac{-3x_0}{4y_0}$

$\therefore$ Eqⁿ of the tangent is

$$y - y_0 = \frac{-3x_0}{4y_0} (x - x_0)$$

$$4yy_0 - 4y_0^2 = -3xx_0 + 3x_0^2$$

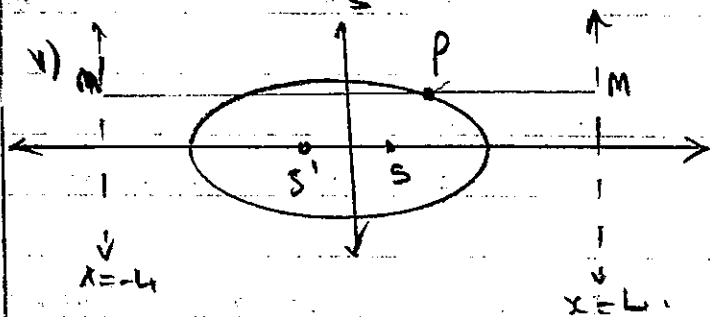
$$3xx_0 + 4yy_0 = 3x_0^2 + 4y_0^2$$

Now $3x_0^2 + 4y_0^2 = 12$

$$\therefore 3xx_0 + 4yy_0 = 12$$

$$\text{i.e. } \frac{xx_0}{4} + \frac{yy_0}{3} = 1$$

(3)



$$\frac{PS}{PM} = e$$

$$\frac{PS'}{PM'} = e$$

$$\therefore PS = ePM \quad \& \quad PS' = ePM'$$

$$PS + PS' = e(PM + PM')$$

$$= \frac{1}{2} \times 8$$

$$= 4$$

which is a constant

(2)

2. (i) If $x = a \sec \theta$
 $\frac{dx}{d\theta} = a \sec \theta \tan \theta$

If $y = b \tan \theta$
 $\frac{dy}{d\theta} = b \sec^2 \theta$

Now $\frac{dy}{dx} = \frac{dy}{d\theta} \times \frac{d\theta}{dx}$
 $= b \sec^2 \theta \times \frac{1}{a \sec \theta \tan \theta}$
 $= \frac{b \sec \theta}{a \tan \theta}$

$\therefore$ the gradient of the normal is
 $= \frac{-a \tan \theta}{b \sec \theta}$

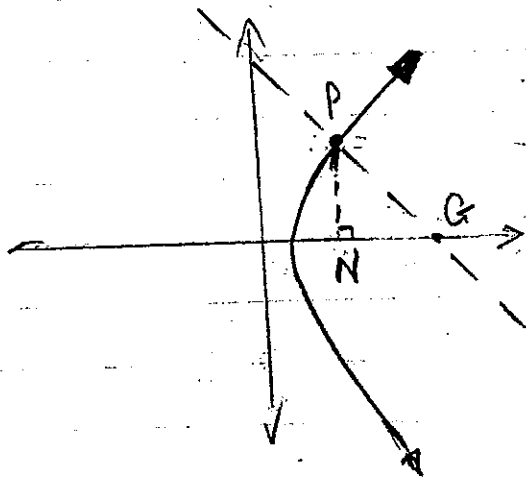
Hence the eqn is

$y - b \tan \theta = \frac{-a \tan \theta}{b \sec \theta} (x - a \sec \theta)$

i.e. $by \sec \theta - b^2 \sec \theta \tan \theta$
 $= -ax \tan \theta + a^2 \sec \theta \tan \theta$

i.e. $a \tan \theta x + b \sec \theta y = (a^2 + b^2) \sec \theta \tan \theta$

(ii)



When $y=0$,
 $a \tan \theta x = (a^2 + b^2) \sec \theta \tan \theta$

$\therefore x = \frac{(a^2 + b^2)}{a} \sec \theta$

$\therefore G$ is $\left(\frac{(a^2 + b^2)}{a} \sec \theta, 0 \right)$

N is $(a \sec \theta, 0)$

$\therefore GN = \left[\frac{(a^2 + b^2)}{a} - a \right] \sec \theta$

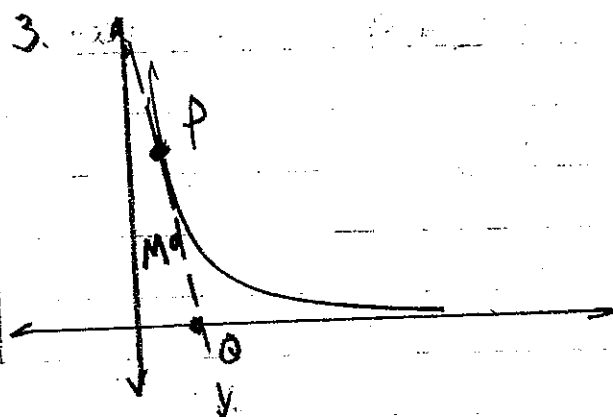
$= \left[\frac{a^2 + b^2 - a^2}{a} \right] \sec \theta$

$= \frac{b^2}{a} \sec \theta$

$ON = a \sec \theta$

$\frac{NG}{ON} = \frac{\frac{b^2}{a} \sec \theta}{a \sec \theta}$
 $= \frac{b^2}{a^2}$

i.e. $NG : ON = b^2 : a^2$



If $xy = c^2$
 $y = c^2 x^{-1}$
 $y' = -c^2 x^{-2}$

When $x = ct$

$$y' = -\frac{c^2}{2ct}$$

$$= -\frac{1}{2t}$$

$\therefore$ Eq of tangent is

$$y - \frac{c}{t} = -\frac{1}{2t}(x - ct)$$

$$2ty - 2ct = -x + ct$$

$$x + 2ty = 2ct$$

When $y = 0$ $x = 2ct$

$$\therefore M \text{ is } \left(\frac{3ct}{2}, \frac{c}{2t} \right)$$

$$\text{If } x = \frac{3ct}{2} \Rightarrow y = \frac{c}{2t}$$

$$\frac{2x}{3c} = t$$

$$\therefore y = \frac{c}{\frac{4x}{3c}}$$

$$y = \frac{3c^2}{4x}$$

$$\text{or } xy = \frac{3c^2}{4}$$

which is another hyperbola

Section III

$$1a) \int_0^2 \frac{2}{\sqrt{x^2+16}} dx$$

$$= 2 \left[\ln(x + \sqrt{x^2+16}) \right]_0^2$$

$$= 2 \left\{ \ln(2+\sqrt{20}) - \ln 4 \right\} \quad (1)$$

$$= 2 \ln\left(\frac{2+\sqrt{5}}{2}\right)$$

$$= 2 \ln\left(\frac{1+\sqrt{5}}{2}\right)$$

$$b) \int \frac{4}{x^2-4x+3} dx = \int \frac{4 dx}{(x-1)(x-3)}$$

$$\text{Let } \frac{4}{(x-1)(x-3)} = \frac{A}{x-1} + \frac{B}{x-3}$$

$$\therefore 4 = A(x-3) + B(x-1)$$

$$\text{If } x=3, \quad 4=2B$$

$$B=2$$

$$\text{If } x=1, \quad 4=-2A$$

$$A=-2$$

$$\therefore I = \int \frac{-2}{x-1} + \frac{2}{x-3} dx$$

$$= -2 \ln(x-1) + 2 \ln(x-3) + C$$

$$= 2 \ln \left| \frac{x-3}{x-1} \right| + C$$

$$c) \int_4^{12} \frac{dx}{\sqrt{x}(4+x)}$$

$$\text{Let } u = \sqrt{x} \text{ or } x = u^2$$

$$du = \frac{1}{2} x^{-1/2} dx$$

$$= \frac{1}{2\sqrt{x}} dx$$

$$\text{If } x=12, \quad u=2\sqrt{3}$$

$$x=4, \quad u=2$$

$$I = 2 \int_4^{12} \frac{dx}{2\sqrt{x}(4+x)}$$

$$= 2 \int_2^{2\sqrt{3}} \frac{du}{u^2+4u}$$

$$= \cancel{2} \left[\frac{1}{\cancel{2}} \tan^{-1} \frac{u}{2} \right]_{-\sqrt{3}}^{2\sqrt{3}}$$

$$= \tan^{-1} \sqrt{3} - \tan^{-1} 1$$

$$= \frac{\pi}{3} - \frac{\pi}{4}$$

$$= \frac{\pi}{12}$$

(4)

d) $\int \frac{dx}{1+\sin x}$ Let $t = \tan \frac{x}{2}$

$$dx = \frac{2}{1+t^2} dt$$

$$\sin x = \frac{2t}{1+t^2}$$

$$\therefore I = \int \frac{1}{1 + \frac{2t}{1+t^2}} \cdot \frac{2}{1+t^2} dt$$

$$= \int \frac{1}{\frac{1+t^2+2t}{1+t^2}} \cdot \frac{2}{1+t^2} dt$$

$$= 2 \int \frac{1}{t^2+2t+1} dt$$

$$= 2 \int \frac{1}{(t+1)^2} dt$$

(3)

e) $\int x e^{3x} dx$ $u=x, v'=e^{3x}$
 $u'=1, v=\frac{1}{3}e^{3x}$

$$I = \frac{x}{3} e^{3x} - \int \frac{1}{3} e^{3x} dx$$

$$= \frac{x}{3} e^{3x} - \frac{1}{9} e^{3x} + c$$

$$= \frac{3xe^{3x} - e^{3x}}{9} + c$$

$$= \frac{e^{3x}(3x-1)}{9} + c \quad (3)$$

2. $x^2 + xy + 4y^2 = 4$

Diff w.r.t x

$$2x + y + x \cdot y' + 8y \cdot y' = 0$$

$$y'(x+8y) = -2x-y$$

$$y' = \frac{-2x-y}{x+8y}$$

At $x=1, y=-1$ $y' = \frac{-2+1}{1-8}$

$$= \frac{-1}{-7} = \frac{1}{7}$$

(3)

$\therefore$ the eqn is

$$y+1 = \frac{1}{7}(x-1)$$

$$7y+7 = x-1$$

$$0 = x-7y-8$$

$$\underline{x-7y-8=0}$$